Crane Payload-Sway Forecasting with Delayed Angle Measurements

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Abstract

When a crane sway measurement arrives late, more recent support-motion measurements may already be available. We study forecasting under this timing difference on 33 published laboratory tower-crane records. The forecaster combines quadratic completion of the missing angle samples with a bank of length-calibrated modal forecasts that start from the latest measured angle. A fixed age-dependent rule blends the two. The forecaster and a matched autoregressive reference with exogenous inputs (ARX) receive the same available histories and angle age. On internal validation, mean absolute angular error was 1.575° for the mixture and 1.758° for the reference under a score that gives equal weight to low and high recent support motion. On seven previously examined records, the mixture had lower mean error in 62 of 63 age–horizon combinations at each of two noise levels. At an angle age of 250 ms and a forecast horizon of 100 ms, its error was higher without added noise, 0.700° versus 0.653°. Overstating the age by 20 ms raised mixture error from 0.581° to 0.869° across five common true ages. Independent validation of the dated forecaster on newly acquired physical records is pending.

Keywords crane payload sway, sensor age, modal forecasting, cable length, autoregressive models

1 Introduction

A sway forecast has two relevant times when the angle measurement arrives late. The latest received angle describes the payload at its measurement time. The forecast describes the payload at a specified horizon after the current time, when newer information about trolley position, jib angle or cable length may already be available. Completing the missing samples allows an existing forecaster to use the current time as its origin. A second approach fits the oscillation state to the dated angle history and propagates it over the intervening motion. These approaches can behave differently at short horizons and when the stated age is wrong.

Earlier crane studies have used cable length and payload mass in data-driven predictors [1], physics-informed neural networks for travelling mobile cranes [2], and observers that compensate for sway-sensor delay [3]. Experimental results from model predictive control on other rigs use different inputs and targets [4, 5]. Here we fit the reference on the same laboratory dataset and evaluate it at the same forecast origins and targets as the mixture.

The present work extends an earlier angle-only analysis of forecast horizons and cable-length extrapolation. It tests a mixture of two causal forecasting approaches when angle age is known. It then asks where the mixture loses accuracy as age, horizon and the accuracy of the supplied age change.

2 Data and forecast protocol

2.1 Laboratory records

The Inteco tower-crane dataset contains 33 laboratory records at 100 Hz, each with 9,901 samples spanning 99 s [6, 7]. The records cover trolley motion, jib rotation and simultaneous operation, with variation in cable length and payload configuration. The nominal sway encoder increment is about 0.0879°. Table 1 gives the record split. Trolley motion is scored on one sway axis and the other families on both axes, giving five family–axis cases and 12 replay record–axis pairs. All 33

recordings had been examined before this experiment. Here *replay* denotes retrospective scoring on the records kept out of fitting and selection in this experiment.

Table 1: Inteco record IDs and sway axes in the dated-angle experiment.

Motion family	Axes	Fit	Validation	Replay
Trolley	θ	1, 2, 3, 5–8, 10–12	4	9, 13
Jib rotation	θ, β	14–16, 18–20	17	21–23
Simultaneous	θ, β	24–27, 29–31	28	32, 33

2.2 Available measurements and targets

At forecast origin t, the latest angle sample was acquired at $t - a$, where a is a known integer multiple of 10 ms. The models receive 401 angle samples ending at $t - a$ and 401 samples of cable length, support position and native drive-reference channels ending at t. Each history spans 4 s. The support channels are trolley position x, jib angle J, or both. The native references x_r and J_r retain the scale of the source export. Their physical actuator units are unverified.

The targets are recorded angles at $t+h$ for $h \in \{0.10, 0.20, 0.35, 0.50, 0.75, 1.00, 1.50\}$ s. Thus the 100 ms target at an angle age of 250 ms lies 350 ms after the last measured angle. Future measured support motion, cable length and sway are excluded from the inputs. Validation and replay origins start at 4.25 s and advance in 1 s steps, preserving the full history at every age without padding the start of a record.

Age and noise are imposed on stored inputs. Validation uses ages of 0, 20, 50, 100, 150 and 200 ms. Replay adds ages of 30, 70 and 250 ms. Each setting uses either the recorded angle history or that history plus seeded uniform jitter of up to one encoder increment in either direction. The recorded target angles and support channels are shared across methods. The condition without added noise retains the original encoder quantization.

3 Forecast methods

3.1 Completing the angle history

The first component fits a quadratic to the last 0.25 s of available angle history. It extrapolates over the missing interval and passes the completed history to a previously fitted forecaster. That forecaster combines autoregressive and modal forecasts using recent support motion and forecast errors on already observed angles. The range of support motion over the past 4 s identifies low-motion packets, using thresholds of 2 mm for trolley position and 2 mrad for jib angle. In these packets the component retains its previous low-motion rule. We also report this component separately as quadratic completion.

3.2 Forecasting from the dated angle

The second component uses six modal experts. They fit a constant offset and oscillatory modes to the last 2.5 s of observed angle history, sampled every 50 ms for the fit. The frequency search is scaled by the nominal frequency $\sqrt{g/\ell_{\text{eff}}}$, with a positive effective length ℓ_{eff} calibrated on fitting records. Four experts align past support measurements to the angle history and use measured support motion through the sensor gap. Beyond t they assume either held or linearly continued support position. Each continuation has one expert with fixed coupling and one with a local regularized update of the fitted coupling. The other two experts describe free motion, one with constant length and one with measured length through the gap followed by held length.

In the changing-length expert, a modal coordinate z_j follows the local approximation

$$\ddot{z}_j + \left(d_j + 2\frac{\dot{\ell}}{\ell}\right)\dot{z}_j + \kappa_j \frac{\ell(t-a)}{\ell} z_j = 0, \tag{1}$$

where d_j and κ_j are modal damping and stiffness at measurement time, and ℓ is effective length. The length-rate term follows the small-angle variable-length pendulum equation [8, 9]. The empirical modes approximate sway along one measured axis at a time.

The bank assigns weights using six forecasts from a packet ending one second earlier. All six targets are observed by t, with the last at $t-a$. The earlier packet uses its own assumed continuation of support motion. Let E contain its six prediction errors for each expert and w_0 be the prior weights. The bank solves

$$\min_w \frac{1}{6}\|Ew\|_2^2 + \lambda\|w - w_0\|_2^2, \qquad w_j \geq 0, \quad \sum_{j=1}^{6} w_j = 1, \tag{2}$$

where $\lambda = \|E\|_F^2/36 + q^2/12$ and q is the encoder increment in the angle units used for fitting. The prior assigns half the weight to each of the two free-motion experts in low support motion. In higher support motion each free expert receives a quarter and each forced expert receives an eighth. The fitted weights apply to the current forecasts at all seven horizons.

3.3 Age-dependent mixture and ARX reference

The selected mixture uses

$$\widehat{\theta}(t+h) = (1-\alpha)\widehat{\theta}_{\text{completion}}(t+h) + \alpha\widehat{\theta}_{\text{bank}}(t+h), \qquad \alpha = \frac{a}{a + 100\,\text{ms}}. \tag{3}$$

At age zero it returns the original synchronous forecast. Internal validation selected this mixture from 21 candidates, including standalone modal forecasts, alternative bank regularization and other blends. The selection, identified as `mix_bank_10` in the repository, was fixed before replay.

The dense ARX reference includes angle-only and input–output feature sets, length-dependent coefficients, direct forecasts and corrections to modal forecasts. Its pool retains 3 s and 4 s history variants and alternative ways to handle missing angle samples. The expanded pool includes 120 new fits and offers 304 options per family–axis case, including some identical predictions. The original dense fits and the new age-augmented fits use 0.25 s steps between training origins. The pipelines have comparable access to fitting records, channels and dense origin support. Their transformations and objectives differ.

The reference is selected on validation for each of five cases and six fitting ages, averaging both noise levels. At other ages it uses the nearest fitting age, resolving ties toward the smaller age. The expanded reference was fitted after the mixture was selected. The mixture and expanded reference were held fixed during their seven-record comparison.

4 Results

4.1 Internal validation and replay averages

We measure mean absolute error (MAE) in degrees relative to the recorded encoder angle. The validation score gives equal weight to five family–axis cases, six ages and two noise conditions. Within each case the score weights low and high recent support motion equally. The mixture scores 1.575° and dense ARX scores 1.758°, a 10.4% reduction on the validation data used in development. The mixture has lower validation error in all five cases. The earlier, narrower ARX pool scored 1.795° on validation. Its all-origin replay error was lower than that of the expanded ARX at ages from 50 to 250 ms. Both references are retained in the archived tables.

For the all-origin replay summary, we first average over origins and horizons within each record and axis, then over axes within records, and then over the seven physical records. Without added

noise, mixture MAE rises from 0.525° at zero age to 0.742° at 250 ms. Dense ARX rises from 1.071° to 1.316°. The direct bank alone has lower average error than the mixture at ages 200 and 250 ms. At 250 ms the bank's MAE is 0.692°.

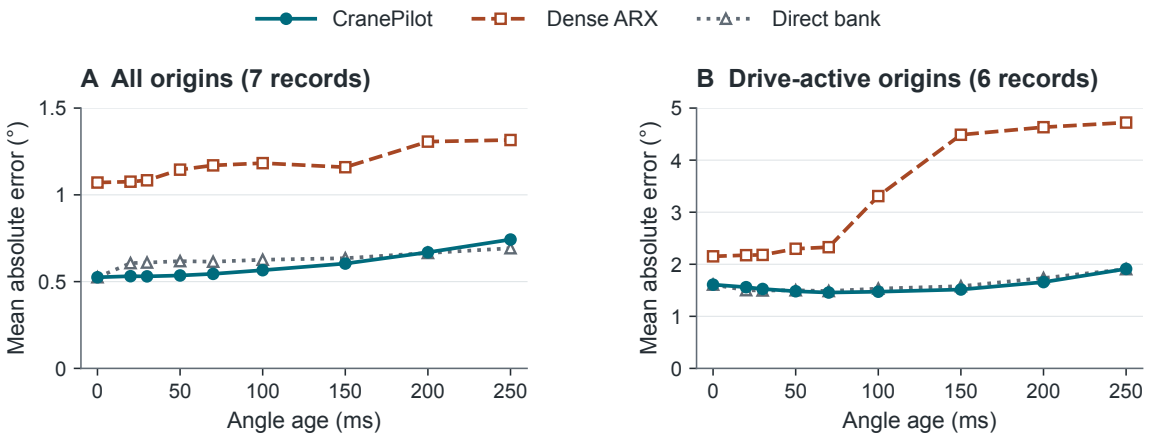


Figure 1: Replay MAE by angle age without added noise. CranePilot denotes the selected mixture. The all-origin comparison covers seven records and 12 record–axis pairs. The drive-active subset covers six records and ten record–axis pairs with active origins. The panels have different vertical scales.

The drive-active subset in Fig. 1 includes support motion with or without concurrent hoisting. Its low-motion complement can include hoisting. Some record–axis groups contain only two or three active origins. Repeating ages, noise conditions and horizons on these records does not create additional independent trials.

4.2 Where the short forecast loses accuracy

Figure 2 shows the horizon curves at zero and 250 ms, and Fig. 3 shows the complete grid. At each noise level, the mixture has lower mean error than dense ARX in 62 of 63 age–horizon cells. At the exception, a 100 ms horizon with 250 ms angle age, the mixture scores 0.700° and dense ARX 0.653° without added noise. With jitter, the corresponding errors are 0.702° and 0.660°. Dense ARX wins on six of the seven physical records in this cell at each noise level.

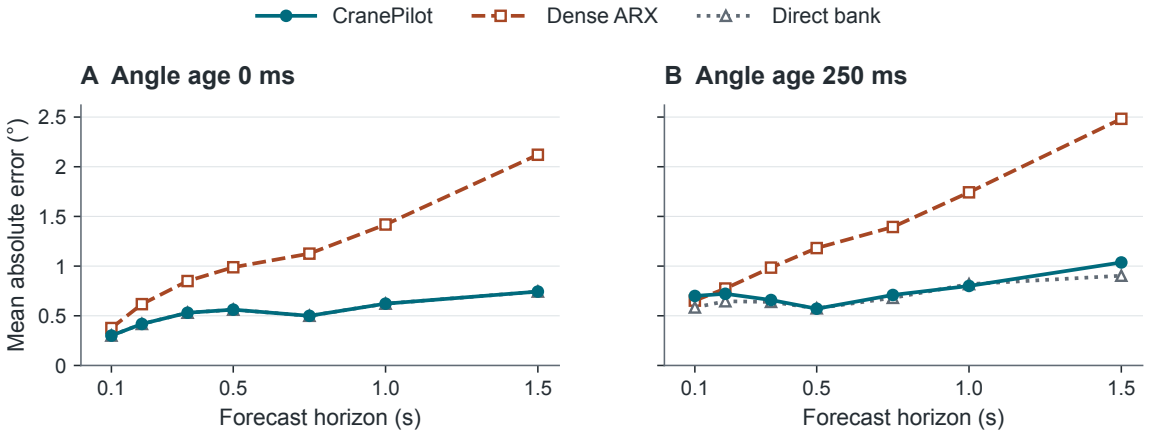


Figure 2: MAE by forecast horizon at angle ages of 0 and 250 ms without added noise. Panels use equal physical-record weights and a common error scale. At 250 ms age the mixture loses to dense ARX at the shortest horizon and wins at the six longer horizons.

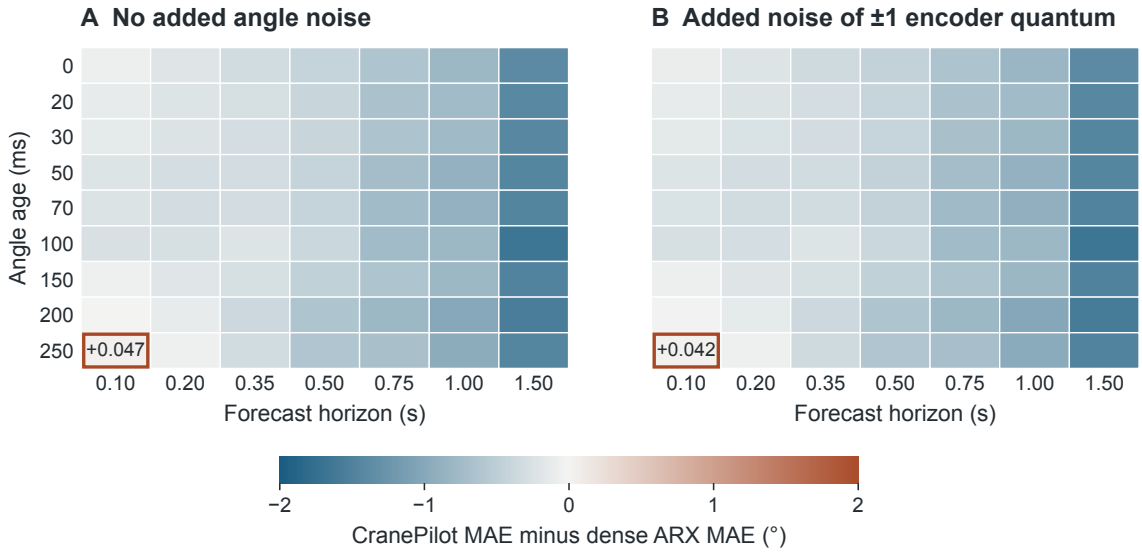


Figure 3: Mixture MAE minus dense ARX MAE on the full replay grid. Negative values favour the mixture. Each panel contains nine ages and seven horizons on the same scale. The outlined cell marks the short-horizon loss in both noise conditions.

4.3 Sensitivity to the reported measurement age

A separate replay without added noise changes the reported age by −20, −10, 0, +10 or +20 ms, retaining combinations within the 0–250 ms input range. The actual histories and targets remain fixed. Five true ages (20, 50, 100, 150 and 200 ms) admit every offset. On this common support, the mixture's mean error increases from 0.581° with correct age to 0.869° with a +20 ms offset, an increase of 49.5%. Dense ARX increases from 1.174° to 1.474° (Fig. 4).

The mixture retains lower MAE averaged over horizons in all 31 admissible age–offset combinations. At a true age of 250 ms and a 100 ms horizon, its mean error remains higher at each admissible offset (−20, −10 and 0 ms).

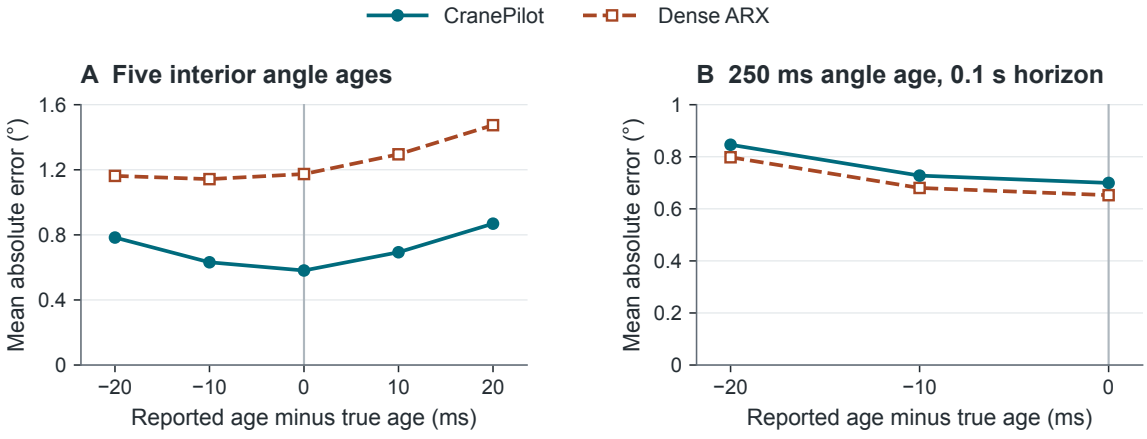


Figure 4: Effect of changing only the stated angle age. A positive offset overstates age. The common-age average uses the same five true ages at every offset. At 250 ms, positive offsets exceed the input range.

5 Discussion

The age-dependent blend lowers retrospective MAE across most of the tested grid. The 100 ms loss at 250 ms age shows why an average across horizons is insufficient for choosing a model for a short-horizon task. The direct bank also outperforms the blend at the longest ages when averaging over horizons. These cases can be tested prospectively with the same frozen models before changing the age weighting.

The target is the original encoder angle, whose resolution matters when interpreting small differences in degrees. The records come from one laboratory apparatus, and cable length, payload and command profiles can vary together. The imposed age, jitter and constant timestamp offsets leave the observed commands and crane trajectory fixed. Real arrival delays, irregular sampling and the response to newly selected control inputs require separate evidence. The native reference scale and clock alignment also need verification on a new rig.

In an earlier transfer experiment on four held-out camera records from a different rotary crane [10], the older CranePilot angular model with geometric and frequency adaptation scored 0.895° versus 0.266° for the geometric reference and 0.038° for a locally fitted ARX residual. That experiment used a different measurement rate and information set. The dated model reported here has yet to be tested on that task or on independently acquired records that meet its input contract.

5.1 Independent confirmation

The prospective protocol fixes both models, the origin and horizon definitions, and the score before evaluation. It requires at least five declared conditions and five independent physical clusters. The primary score weights conditions, trials within conditions, axes within trials, and horizons and origins within axes. A positive result requires at least 10% lower MAE, a strict majority of condition wins and a negative upper 97.5th percentile for a paired bootstrap difference that resamples whole clusters. Suitable new trials have not yet been received.

6 Conclusion

Combining angle-history completion with dated modal forecasts lowers mean sway prediction error on most of the tested Inteco grid. The short-horizon loss at the longest angle age and the sensitivity to a 20 ms age error identify two cases to retain in independent validation. The current evidence concerns forecast accuracy on previously recorded trajectories. Independent physical validation and closed-loop control evaluation of the dated forecaster remain open.

Data and code availability

The Inteco data are available from Mendeley Data, version 2 [7]. The CranePilot repository contains the frozen bundles, record splits, validation choices, aggregate replay tables and figure script. The evidence is tied to research snapshot `f3107fe`. The timestamp audit verified 84 archived prediction files and matched all 15,792 correct-age forecast vectors to the preceding replay. The snapshot passed 319 software tests.

Acknowledgments and disclosure

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A Earlier CranePilot experiments

In the earlier angle-only Python benchmark on four trolley records, a sensitivity-projected neural residual reduced MAE at 100 ms from 0.300° to 0.219° relative to a calibrated oscillator. At 1 s it increased MAE from 0.758° to 0.907°. The 41-sample history and record split differ from the dated-angle task. In a separate command-conditioned experiment, a passive-memory model satisfied a discrete storage balance but did not consistently outperform a Gaussian-process residual given the same input information.

A subsequent command-conditioned temporal comparison retained the driven innovation model on the primary split (0.5005° MAE). On long-cable transfer, the validation-selected temporal candidate reduced MAE from 0.513° to 0.497° across all origins, while increasing the active-origin error from 0.932° to 0.968°. The selected short-cable candidate increased mean error from 0.432° to 0.439°. These experiments used planned command inputs and a different split.

The original control pilot compared nominal and geometrically calibrated model predictive control in 16 simulated contexts with synthetic plant mismatch and assumed state feedback. That pilot evaluated separate controllers. The current dated forecaster has yet to be evaluated in a feedback loop.

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